

Equivariant Systems Theory and Observer Design for Autonomous Systems

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Outline

- 1 Matrix ODEs
- 2 ODE with model plus correction
- 3 Fréchet derivatives
- 4 Gradients

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What is a matrix ODE?

scalar ODE

$$\dot{x}(t) = f(t, x(t)), \quad x(0) = x_0 \in \mathbb{R}$$

vector ODE

$$\underbrace{\begin{pmatrix} \dot{x}_1(t) \\ \vdots \\ \dot{x}_n(t) \end{pmatrix}}_{\dot{x}(t)} = \underbrace{\begin{pmatrix} f_1(t, x_1(t), \dots, x_n(t)) \\ \vdots \\ f_n(t, x_1(t), \dots, x_n(t)) \end{pmatrix}}_{f(t, x(t))}, \quad x(0) = x_0 \in \mathbb{R}^n$$

matrix ODE

$$\underbrace{\begin{pmatrix} \dot{x}_{11}(t) & \dots & \dot{x}_{1m}(t) \\ \vdots & & \vdots \\ \dot{x}_{n1}(t) & \dots & \dot{x}_{nm}(t) \end{pmatrix}}_{\dot{X}(t)} = \underbrace{\begin{pmatrix} f_{11}(t, x_{ij}(t)) & \dots & f_{1m}(t, x_{ij}(t)) \\ \vdots & & \vdots \\ f_{1n}(t, x_{ij}(t)) & \dots & f_{nm}(t, x_{ij}(t)) \end{pmatrix}}_{F(t, X(t))}, \quad X(0) = X_0 \in \mathbb{R}^{n \times m}$$

Matrix manifolds

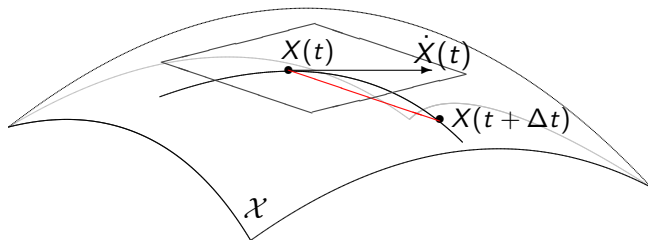
In many cases of interest the solutions of matrix ODEs are not arbitrary matrices but matrices with a particular structure, e.g. $R(t) \in \text{SO}(3)$ or $X(t) \in \mathbb{R}^{n \times m}$ with full rank.

The set of matrices with this structure then forms a *smooth submanifold* $\mathcal{X}$ of $\mathbb{R}^{n \times m}$ and the right hand side of the ODE always lies in the *tangent space* at $X(t)$ to this submanifold.

The tangent space at $X(t)$ to a submanifold is a subspace of $\mathbb{R}^{n \times m}$ and can be explicitly computed, e.g. for $\text{SO}(3)$, $\text{SE}(3)$, etc.

The meaning of the right hand side

$$\dot{X}(t) = \lim_{\Delta t \rightarrow 0} \frac{X(t + \Delta t) - X(t)}{\Delta t}$$



$$\dot{X} = F(t, X)$$

Example: Attitude kinematics

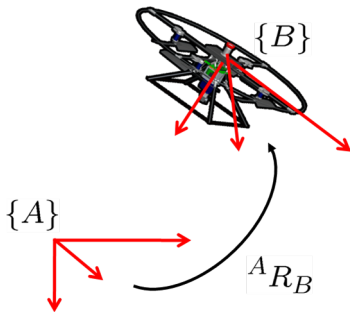
Attitude can be represented as rotation matrices in the special orthogonal group

$$\text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det(R) = 1\}$$

$$\dot{R} = R\Omega_{\times}$$

$\Omega \in \{B\}$ is the rigid body angular velocity measured in the body-fixed-frame.

$$\Omega_{\times} = \begin{pmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{pmatrix}$$



Numerical integration of matrix ODEs

When computing a numerical solution to a matrix ODE, it is quite important to try to preserve the structure of the matrix solutions. E.g., if we integrate attitude kinematics, we want to make sure that our approximate solutions consist of rotation matrices.

A standard Euler or Runge-Kutta method *will not* preserve this structure. People often “solve” this problem by projecting the numerical solution back onto the submanifold. There is no guarantee that such a scheme is physically meaningful.

A principled alternative are so called *Lie group integrators*. We will demonstrate a Lie group Euler method in a future session.

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What does an observer ODE look like

For a kinematic system

$$\dot{X} = XU$$

on a matrix Lie group $\mathbf{G}$ (a special type of matrix manifold) a typical observer looks like

$$\dot{\hat{X}} = \hat{X}U + \Delta\hat{X}$$

where both the *internal model* $\hat{X}U$ and the *correction term* $\Delta\hat{X}$ are tangent vectors at $\hat{X} \in \mathbf{G}$.

U and Δ are then elements of the Lie algebra $\mathfrak{g}$.

Correction terms for observers

The idea behind finding a *good* correction term $\Delta\hat{X}$ is to try and find a *descent direction* for a *cost function* that measures how far $\hat{X}$ is from X .

Possibly the simplest way of finding a descent direction is to compute a *gradient* of the cost function. To do this on a matrix manifold, we need to first understand what a *differential* is.

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Reminder: Lyapunov method for stability analysis

When analyzing stability of a stationary point of

$$\dot{x} = f(t, x)$$

with the help of a Lyapunov function

$$\mathcal{L}(t, x)$$

we compute

$$\dot{\mathcal{L}} = \frac{\partial \mathcal{L}}{\partial x}(t, x(t)) \cdot \dot{x}(t) + \frac{\partial \mathcal{L}}{\partial t}(t, x(t)),$$

the time derivative of $\mathcal{L}$ along solutions.

matrix case: $\mathcal{L}(t, X) \rightarrow \text{"}\frac{\partial \mathcal{L}}{\partial X}\text{"}???$

Directional derivative in $\mathbb{R}^n$

Multivariable calculus

$$\frac{\partial f}{\partial x_i}(x_1, \dots, x_i, \dots, x_n) = \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_i + h, \dots, x_n) - f(x_1, \dots, x_i, \dots, x_n)}{h}$$

$$\begin{aligned} f(x_1, \dots, x_i + h_i, \dots, x_n) &= f(x_1, \dots, x_i, \dots, x_n) \\ &\quad + \frac{\partial f}{\partial x_i}(x_1, \dots, x_i, \dots, x_n) \cdot h_i \\ &\quad + O(h_i^2) \end{aligned}$$

$$f(x + h) = f(x) + Df(x)[h] + O(\|h\|^2)$$

Directional derivative in $\mathbb{R}^n$

$$f(x+h) = f(x) + Df(x)[h] + O(\|h^2\|)$$

We have

$$\begin{aligned} Df(x)[h] &= \begin{pmatrix} \frac{\partial f}{\partial x_1}(x_1, \dots, x_n) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x_1, \dots, x_n) \end{pmatrix}^\top h \\ &= \lim_{t \rightarrow 0} \frac{f(x+th) - f(x)}{t} \\ &= \left. \frac{d}{dt} \right|_{t=0} f(x+th) \end{aligned}$$

Directional derivative in $\mathbb{R}^n$

For maps

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

we get the *Jacobian*

$$\begin{aligned} Df(x)[h] &= \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix} h \\ &= \frac{d}{dt} \Big|_{t=0} f(x + th) \end{aligned}$$

The Fréchet derivative on matrix manifolds

For functions on matrix manifolds

$$F: \mathcal{X} \rightarrow \mathbb{R}$$

we can think of F as defined on all of $R^{n \times m}$ and then choose H in

$$DF(X)[H] = \left. \frac{d}{dt} \right|_{t=0} F(X + tH)$$

as an element of the tangent space at X to $\mathcal{X}$.

All this extends to maps $F: \mathcal{X} \rightarrow \mathcal{Y}$ between matrix manifolds with

$$DF(X): T_X \mathcal{X} \rightarrow T_{F(X)} \mathcal{Y}$$

a *linear* map between tangent spaces, the *differential* of F at X .

Example: $F(X) = X$

Compute $DF(X)[H] = ???$

Example: $F(X) = X$

We find $DF(X)[H] = H$, i.e. $DF(X) = \text{id}_{T_X \mathcal{X}}$.

Example: $F(X) = X^T X$

Compute $DF(X)[H] = ???$ for $X \in \mathbb{R}^{n \times n}$

Example: $F(X) = X^\top X$

We find $DF(X)[H] = X^\top H + H^\top X$.

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Gradients in $\mathbb{R}^n$

Given an inner product $\langle x, y \rangle$ on $\mathbb{R}^n$ we can define the *gradient* of a function

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

with respect to $\langle \cdot, \cdot \rangle$ by the requirement

$$Df(x)[h] = \langle \text{grad } f(x), h \rangle$$

for all h in $\mathbb{R}^n$ (technically $h \in T_x \mathbb{R}^n$).

What do we get for $\langle x, y \rangle = x^\top y$?

Riemannian metrics

A *Riemannian metric* is a family of inner products, one on each tangent space of a manifold, that varies smoothly with the base point.

The tangent spaces to matrix manifolds are subspaces of $\mathbb{R}^{n \times m}$, so we can for example use the *Frobenius inner product*

$$\langle A, B \rangle = \text{tr} \left(A^\top B \right)$$

Compute this for two elements of $T_R \text{SO}(3)$...

Gradients on matrix manifolds

The definition of a gradient is now the same as for the case of $\mathbb{R}^n$:

$$DF(X)[H] = \langle \text{grad } F(X), H \rangle$$

for all $H \in T_X \mathcal{X}$ but with the additional condition that

$$\text{grad } F(X) \in T_X \mathcal{X}.$$

A gradient on $\mathbb{R}^{n \times n}$

Let's compute the gradient of the function

$$\begin{aligned} F: \mathbb{R}^{n \times n} &\rightarrow \mathbb{R} \\ X &\mapsto F(X) = \text{tr}(I - X) \end{aligned}$$

with respect to the Frobenius inner product.

A gradient on $\text{SO}(3)$

Let's compute the gradient of the function

$$\begin{aligned} F: \text{SO}(3) &\rightarrow \mathbb{R} \\ X &\mapsto F(X) = \text{tr}(I - X) \end{aligned}$$

with respect to the Frobenius inner product.